

TULUN, Russian Federation

WMO: 305040

Lat: 54.562N

Lon: 100.579E

Elev: 522

StdP: 95.21

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.8	-32.6	-38.1	0.1	-35.7	-35.0	0.1	-32.4	6.2	-13.3	5.4	-14.8	0.2	130	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	28.5	17.8	26.8	17.0	25.1	16.3	19.3	25.6	18.5	24.4	17.6	23.0	2.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	13.2	21.3	16.3	12.4	20.6	15.4	11.7	20.0	57.7	26.0	54.6	24.5	51.7	23.1	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.2	4.4	-37.6	31.7	4.6	2.0	-41.0	33.2	-43.7	34.3	-46.2	35.4	-49.6	36.9		
(5)			-37.7	21.0	4.6	1.3	-41.0	22.0	-43.7	22.7	-46.3	23.5	-49.6	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-19.1	-15.4	-7.0	2.2	9.0	15.5	18.1	15.6	8.6	0.5	-10.0	-17.2
	DBStd	14.13	7.02	6.93	6.42	5.03	5.06	4.05	2.93	3.59	4.47	5.51	7.18	7.38
	HDD10.0	4284	901	712	528	239	82	5	0	3	79	296	598	842
	HDD18.3	6702	1159	945	786	484	291	101	41	99	294	553	848	1101
	CDD10.0	689	0	0	0	4	51	171	249	176	36	1	0	0
	CDD18.3	66	0	0	0	0	2	17	32	14	1	0	0	0
	CDH23.3	781	0	0	0	1	52	235	334	146	12	0	0	0
	CDH26.7	159	0	0	0	0	9	51	71	28	1	0	0	0
(14) Wind	WSAvg	1.8	1.2	1.4	1.9	2.3	2.4	2.0	1.8	1.8	1.8	1.9	1.7	1.3
(15) Precipitation	PrecAvg	416	12	9	7	20	33	56	88	88	44	22	17	17
	PrecMax	575	40	26	25	59	76	132	184	210	84	75	30	39
	PrecMin	272	3	4	0	2	12	24	14	35	14	0	3	5
	PrecStd	71	7	5	5	14	15	23	39	44	18	16	7	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.5	1.8	8.6	20.1	28.1	30.4	31.1	29.5	25.1	17.1	6.5	0.1
		MCWB	-3.1	-2.2	2.6	9.3	15.5	17.9	19.6	18.2	15.0	8.6	2.3	-2.7
	2%	DB	-4.0	-1.3	6.0	16.2	24.7	28.1	28.6	26.9	21.3	13.4	3.6	-3.4
		MCWB	-5.2	-3.8	1.3	7.7	12.9	16.9	18.6	17.6	13.2	6.9	0.6	-4.9
	5%	DB	-7.0	-3.5	3.9	13.2	21.6	26.1	26.8	24.8	18.5	10.5	1.7	-5.5
		MCWB	-8.1	-5.5	0.0	6.0	11.4	16.1	17.9	16.6	11.8	5.5	-0.6	-6.6
	10%	DB	-9.7	-5.9	2.0	10.5	18.7	24.1	25.1	22.7	15.9	8.0	-0.3	-7.7
		MCWB	-10.5	-7.5	-1.3	4.6	9.8	15.2	17.2	15.8	10.7	4.2	-1.9	-8.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-1.5	3.2	10.0	15.9	19.7	21.3	19.6	16.3	9.7	2.9	-2.6
		MCDB	-2.1	1.0	7.5	17.5	26.3	26.0	27.6	26.6	23.8	15.2	5.8	-0.5
	2%	WB	-5.3	-3.7	1.6	8.3	13.8	18.4	19.8	18.3	14.0	7.6	0.9	-4.7
		MCDB	-4.1	-1.4	5.6	15.1	22.8	25.4	25.6	24.4	19.4	12.2	3.0	-3.4
	5%	WB	-8.1	-5.5	0.2	6.6	12.2	17.3	19.0	17.4	12.6	6.0	-0.4	-6.5
		MCDB	-7.0	-3.6	3.5	12.7	20.1	23.7	25.2	23.1	17.4	10.3	1.4	-5.5
	10%	WB	-10.5	-7.5	-1.1	4.8	10.6	16.2	18.2	16.5	11.1	4.4	-1.9	-8.4
		MCDB	-9.7	-5.9	1.9	10.3	17.8	22.4	23.4	21.5	15.1	7.6	-0.2	-7.7
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	10.1	10.7	9.8	11.9	11.8	10.7	10.2	10.0	8.7	7.3	7.0
		MCDBR	7.9	11.1	11.8	14.4	16.6	15.5	14.2	14.2	14.4	13.1	8.2	8.0
	5% WB	MCWBR	7.1	9.0	8.1	8.0	8.0	6.4	5.9	5.9	6.9	7.3	6.1	7.0
		MCDWR	7.7	10.5	10.5	13.5	15.0	12.9	11.8	12.2	12.7	12.0	7.8	7.7
(40) Clear-Sky Solar Irradiance	taub		0.273	0.304	0.337	0.377	0.431	0.419	0.417	0.388	0.344	0.328	0.299	0.274
		taud	2.047	2.029	2.039	2.192	2.096	2.241	2.275	2.339	2.420	2.326	2.152	2.006
	Ebn at Noon		640	753	819	844	813	829	823	827	823	738	606	534
		Edh at Noon	69	101	127	126	148	129	123	109	88	76	63	58
(44) All-Sky Solar Radiation	RadAvg		1.01	2.07	3.71	4.83	5.36	5.97	5.40	4.53	3.15	1.83	0.96	0.69
	RadStd		0.06	0.10	0.15	0.37	0.47	0.34	0.46	0.29	0.29	0.15	0.09	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page